

CE-312 Assignment # 1 (CLO-1)

Classification of Daily Site Conditions

Name: _____

Roll No: _____

Section: _____

Date: _____

Given Data

x = Crew productivity (m^3/hour), y = Equipment utilization (hour/day), Label: 1 = Acceptable, 0 = Unacceptable

<u>Day</u>	<u>x</u>	<u>y</u>	<u>Label</u>
D1	1	5	1
D2	1	3	0
D3	2	3	1
D4	2	1	0
D5	3	2	1
D6	3	1	0
D7	4	2	1
D8	4	1	0
D9	5	1	0
D10	5	2	1

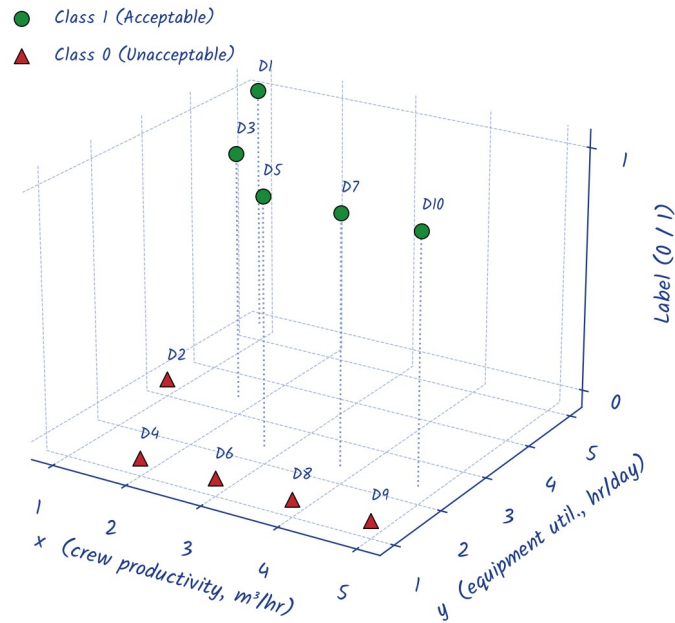
Part (a) Examining the data and 3D plot

First I looked at the data carefully before plotting. My observations are:

- There are 10 days in total. 5 days are Acceptable (D1, D3, D5, D7, D10) and 5 are Unacceptable (D2, D4, D6, D8, D9), so the data is balanced.
- x goes from 1 to 5 and every value of x comes exactly two times, once with label 1 and once with label 0. So x alone can not tell us the class.
- If I compare the two days having the same x , the day with the higher y is always class 1 (e.g. D1 $y=5$ vs D2 $y=3$, D3 $y=3$ vs D4 $y=1$, D9 $y=1$ vs D10 $y=2$). So y is important and both x and y together decide the class.

For the 3D plot I took x on the x -axis, y on the y -axis and the Label (0 or 1) on the z -axis. Green circles are class 1 and red triangles are class 0. Class 1 points are lifted up to $z = 1$ and class 0 points stay on the floor ($z = 0$).

Fig 1: 3D plot of the given dataset



From Fig 1 it is clear that the green points are towards the higher y side of the graph while red points are mostly at $y = 1$ or at low y. There is no simple line on x only that can separate them.

Part (b) Comparing $z = xy$ and $z = -\log(x)$

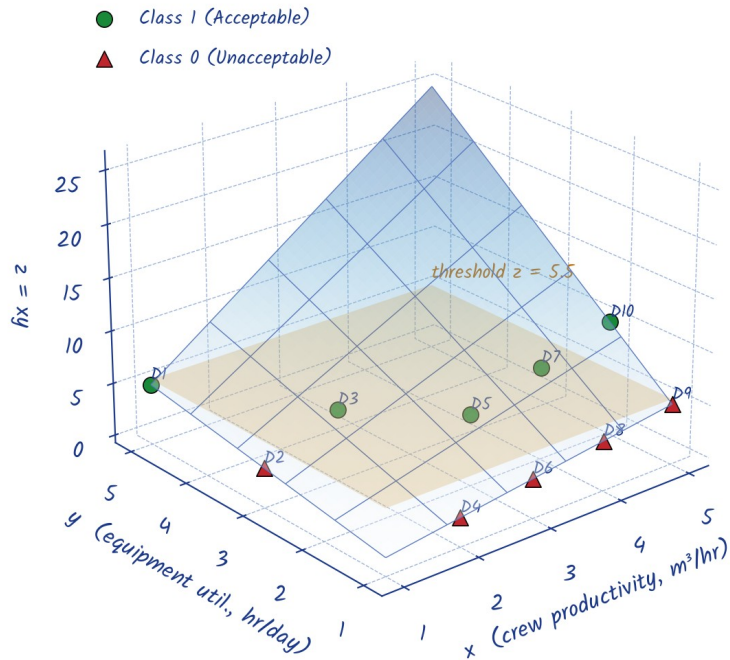
I calculated both functions for every day. (log is taken to base 10. With natural log the numbers change but the conclusion stays same because it still depends only on x.)

<u>Day</u>	<u>x</u>	<u>y</u>	<u>$z = xy$</u>	<u>$z = -\log(x)$</u>	<u>Actual Label</u>
D1	1	5	5	0	1
D2	1	3	3	0	0
D3	2	3	6	-0.301	1
D4	2	1	2	-0.301	0
D5	3	2	6	-0.477	1
D6	3	1	3	-0.477	0
D7	4	2	8	-0.602	1
D8	4	1	4	-0.602	0
D9	5	1	5	-0.699	0
D10	5	2	10	-0.699	1

Function 1: $z = xy$

This surface goes up when x or y increases (it is like a curved sheet, highest at $x = 5, y = 5$). When I put the data points on it (Fig 2), all green points are at $z = 5, 6, 6, 8, 10$ and red points are at $z = 3, 2, 3, 4, 5$. So green points sit higher on the surface and red points sit lower. Only D1 and D9 both have $z = 5$, so they overlap.

Fig 2: $z = xy$ plotted on the dataset



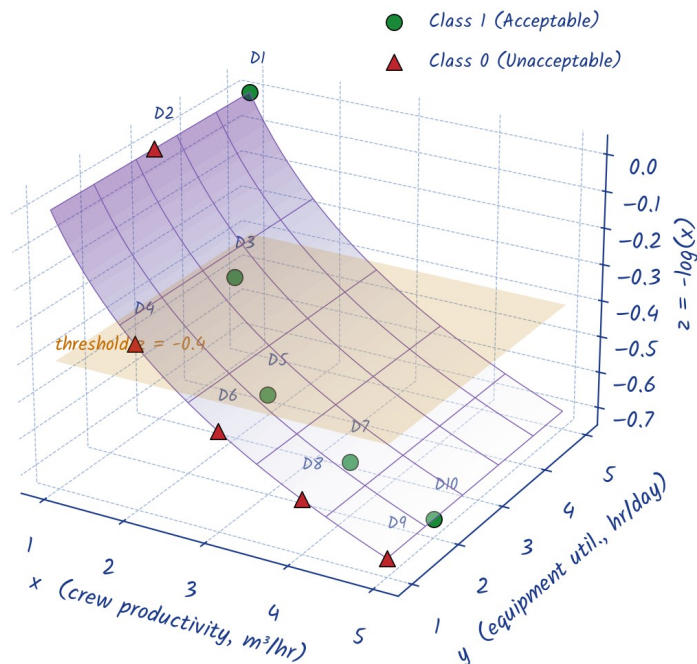
Average z of class 1 = $(5+6+6+8+10)/5 = 7.0$

Average z of class 0 = $(3+2+3+4+5)/5 = 3.4$

Function 2: $z = -\log(x)$

This function only uses x , there is no y in it. So the surface is flat in the y direction and just goes down as x increases (Fig 3). Because each x value has one green and one red point, both points of the pair get the exact same z value (e.g. D1 and D2 both get 0, D3 and D4 both get -0.301).

Fig 3: $z = -\log(x)$ plotted on the dataset



Average z of class 1 = $(0 - 0.301 - 0.477 - 0.602 - 0.699)/5 = -0.416$

Average z of class 0 = $(0 - 0.301 - 0.477 - 0.602 - 0.699)/5 = -0.416$

Both averages are exactly the same, which shows $-\log(x)$ has no ability to separate the two classes.

Comparison:

<u>Point</u>	<u>$z = xy$</u>	<u>$z = -\log(x)$</u>
Variables used	x and y both	only x
Class 1 values	5 to 10 (high)	same as class 0
Class 0 values	2 to 5 (low)	same as class 1
Separation	good (only 1 overlap)	none

So from the plots, $z = xy$ looks much better for this dataset.

Part (c) Method to test the accuracy

Method (as per hint, using a threshold T):

- Step 1: Calculate z for each day using the function.
- Step 2: Pick a threshold T . Rule: if $z > T$ then predicted = 1 (Acceptable), otherwise predicted = 0.
- Step 3: Compare predicted label with the actual label for all 10 days.
- Step 4: Accuracy = (number of correct predictions / total days) $\times$ 100

To choose T , I tried a few values in between the z values and counted the correct ones:

Trying thresholds for $z = xy$

<u>T</u>	<u>Correct</u>	<u>Accuracy</u>
2.5	6 / 10	60 %
3.5	8 / 10	80 %
4.5	9 / 10	90 %
5.5	9 / 10	90 %
6.5	7 / 10	70 %

4.5 and 5.5 both give 90 %. I selected $T = 5.5$ because with this value no Unacceptable day is called Acceptable, which is the safer side for a construction site.

Trying thresholds for $z = -\log(x)$

<u>T</u>	<u>Correct</u>	<u>Accuracy</u>
-0.15	5 / 10	50 %
-0.40	5 / 10	50 %
-0.55	5 / 10	50 %
-0.65	5 / 10	50 %

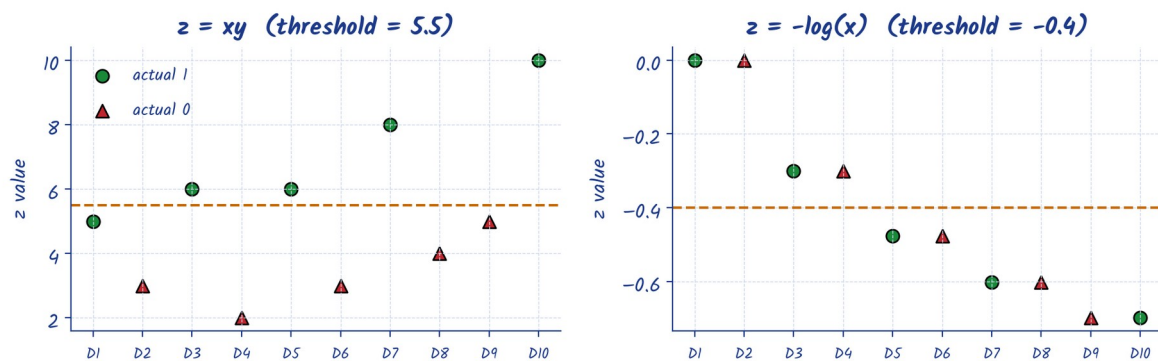
Every threshold gives 50 %. I selected $T = -0.4$ (i.e. predict acceptable when x is 1 or 2).

Checking predicted vs actual (day by day)

(xy uses $T = 5.5$ and $-\log(x)$ uses $T = -0.4$)

Day	Actual	xy	Pred	Result	$-\log(x)$	Pred	Result
D1	1	5	0	wrong	0	1	correct
D2	0	3	0	correct	0	1	wrong
D3	1	6	1	correct	-0.301	1	correct
D4	0	2	0	correct	-0.301	1	wrong
D5	1	6	1	correct	-0.477	0	wrong
D6	0	3	0	correct	-0.477	0	correct
D7	1	8	1	correct	-0.602	0	wrong
D8	0	4	0	correct	-0.602	0	correct
D9	0	5	0	correct	-0.699	0	correct
D10	1	10	1	correct	-0.699	0	wrong

Fig 4: Points above the dashed line are predicted as Class 1



Confusion matrix

	$z = xy$	$z = -\log(x)$
TP (actual 1, pred 1)	4	2
TN (actual 0, pred 0)	5	3
FP (actual 0, pred 1)	0	2
FN (actual 1, pred 0)	1 (D1)	3
Accuracy	$(4+5)/10 = 90 \%$	$(2+3)/10 = 50 \%$

Conclusion

- $z = xy$ gives 90 % accuracy. Only D1 is wrong (its $z = 5$ is same as D9 which is unacceptable).
- $z = -\log(x)$ gives only 50 % accuracy for any threshold, which is same as guessing by tossing a coin. This is because it ignores y (equipment utilization) and the data has both classes at every x .

- So $z = xy$ is the more suitable function to classify the site conditions. It makes sense also, because x times y is like the total work output of crew and equipment together.